

EXERCISE 2

Multimedia Databases (Exercises)

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1: Semantic gap



1: Low-level and High-Level Features

- **Low-level features:**

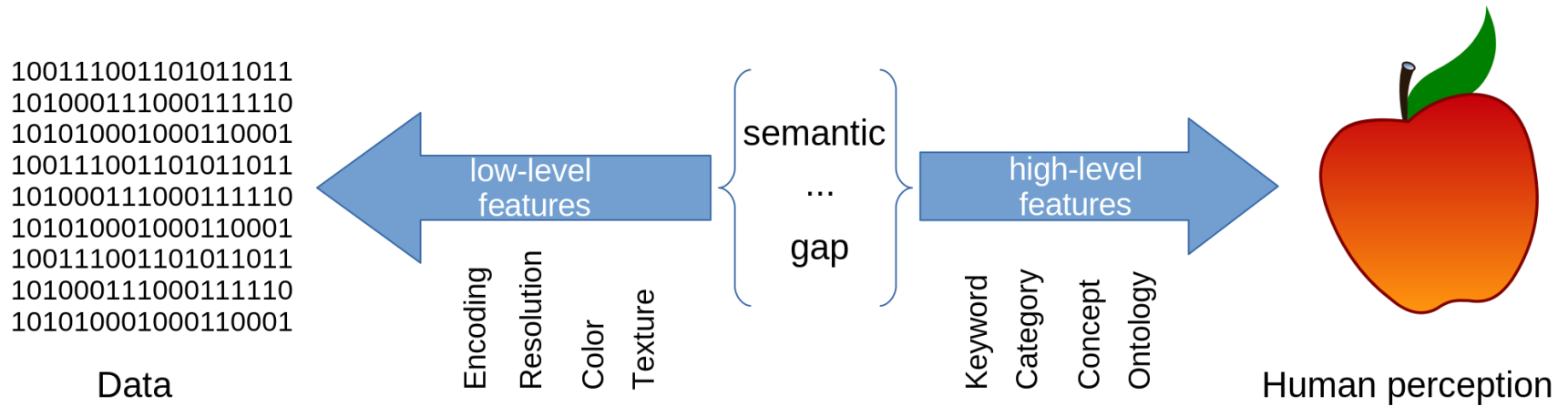
- Examples: Color histogram, pixel intensity, pixel gradient etc.
- Easy to extract, directly from data
- Next to no semantics

- **High-level features:**

- Examples: Winter sports, Person Skiing, snow, mountain
- More semantics
- More difficult to extract



1: Bridging Semantic Gap



1: Low-level and High-Level Features

Example: Image 2 Text BLIP

- Helpfull?
- Flattering?
- Semantics?

Try it out

[Web AI Next.js image to text demo](https://web-ai-demo.vercel.app/demos/image-to-text)

<https://web-ai-demo.vercel.app/demos/image-to-text>



Prefix/question:

Result:

a man with a beard and a beard

1: Low-level and High-Level Features

The image is a medium close-up portrait of a man, centered against a background of indoor greenery.

Subject

- **Person:** A man with a fair complexion, short brown hair, and a well-groomed brown beard and mustache. He is smiling directly at the camera, showing his teeth.
- **Attire:** He is wearing a dark gray or charcoal-colored hooded sweatshirt (hoodie). The hood is down, and the drawstring is visible. Underneath, a dark t-shirt is partially visible.

Background and Composition

- **Background:** Directly behind the man is a large indoor plant with long, narrow, dark green leaves (resembling a Dracaena or similar tropical plant). The background is softly blurred, indicating a shallow depth of field.
- **Lighting:** The lighting is bright and even, likely coming from a large window or soft indoor source, which creates a gentle highlight on his face and hair.
- **Framing:** The shot is framed from the chest up, with the man positioned in the center of the frame.

Atmosphere

The overall mood of the photo is warm, friendly, and casual. The combination of his expression and the natural greenery creates a relaxed, approachable feel.

1: Low-level and High-Level Features

is the same person depicted?



No, these are **not the same person**. While they share some general features, such as eye color and basic face shape, several distinct anatomical differences confirm they are different individuals:

- **Ear Shape:** The shape and "wing" of the ears differ significantly between the two men.
- **Eyebrows:** The second man has a very distinct depigmented patch (vitiligo) on his left eyebrow that is completely absent in the first man.
- **Facial Features:** The structure of the nose and the specific alignment of the teeth (visible in their smiles) are different.
- **Hairline:** The natural hairline and growth patterns are distinct, even accounting for the different styling (casual vs. slicked back).

The first image appears to be a casual personal photo, while the second looks like a professional headshot, likely for a medical or corporate profile.

- **Image 1:** Casual, wearing a **dark gray hoodie**.
- **Image 2:** Formal, appearing to wear a **white collared shirt** (the top of a stethoscope is also partially visible, suggesting a medical professional's headshot).

1: Low-level and High-Level Features

Audio equivalents:

Low-level Features:

- Audiolength
- Bit-/Samplerate
- Frequency spectrum (FT Analyzer)
- dynamic range (lowest vs. Highest Amplitude)
- Signal-to-Noise ratio

High-level Features:

- Speech content
- Gender of the speaker
- Physical features of the audio source
- Emotional connotation



2: MMDB vs Classical Databases

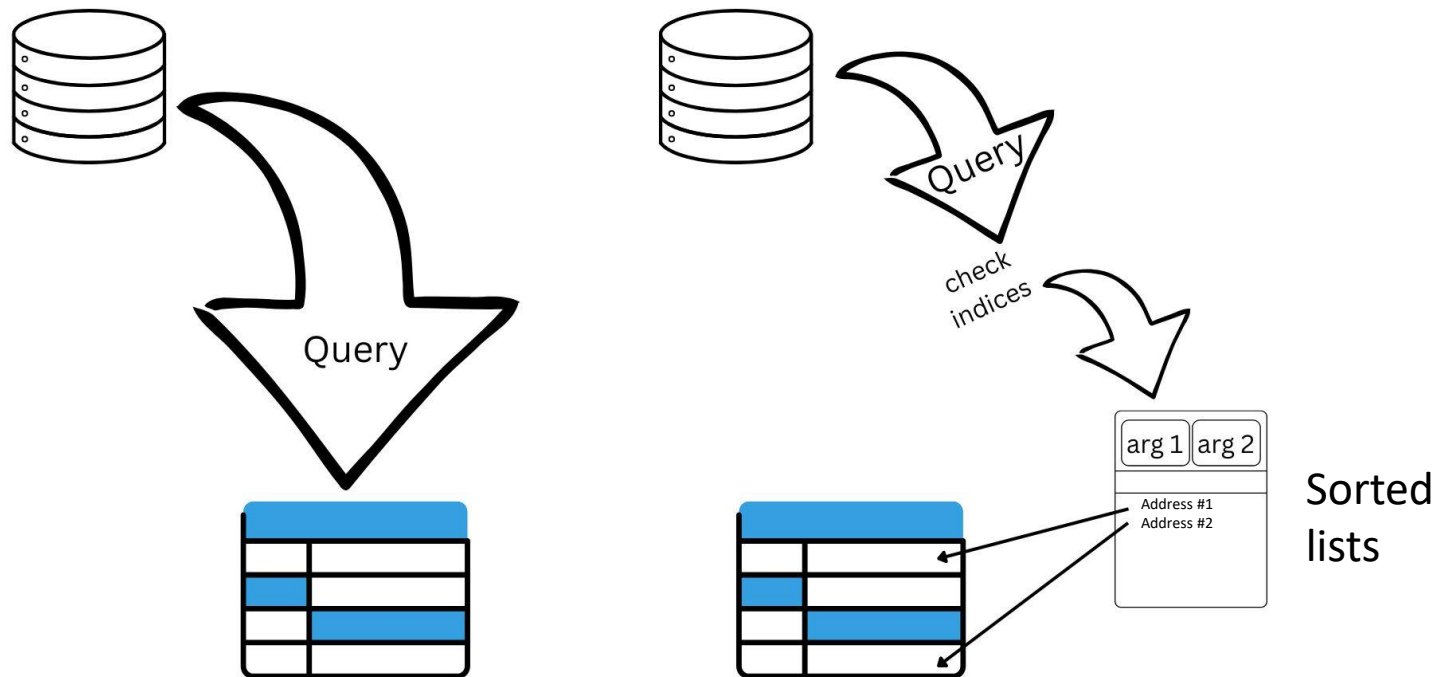
- Data model: Contains unstructured data, no fixed model. Must manage both unstructured and structured metadata.
- Data Volume: MM databases often have huge volumes of data.
- Indexing:
 - Metadata can be multi-dimensional (e.g. color histogram) → classical indexing can not be adopted
 - Keywords(metadata) are by far the predominant method used for indexing multimedia data
 - Automated indexing uses features such as color, shape, texture, spatial information for images. Note, tone, duration for music data.



2: MMDB vs Classical Databases

Questions:

How does database indexing work?



2: MMDB vs Classical Databases

Questions:

Can we build a unique URL for any MM datapoint?

Sure

use a (pseudo-random) string: (Youtube Video ID)

dQw4w9WgXcQ?si=horRpDWKVmtnhO8A

run a hash-algorithm on (parts) of the data

So you can find things you know how to look for ... but searching ...



2: MMDB vs Classical Databases

- Querying and information retrieval:
 - Retrieval algorithms must support content and context-based retrieval
 - Should offer support for spatial and temporal queries
 - Querying by examples
 - Flexible querying using fuzzy predicates.



2: MMDB vs Classical Databases

Video platform:

- Account:
 - Structured data model
 - SQL ... fixed matching and indexing
 - Embedded links to other functionalities
 - Video history
 - Account functionalities
 - Saved information



2: MMDB vs Classical Databases

Video platform:

- Comment section:
 - Structured data model
 - SQL ... fixed matching and indexing
 - Is embedded within a semi structured context
 - Can break for any number of reasons
 - ✓ Account gets deleted
 - ✓ Video gets deleted
 - ✓ Metadata structure changes
 - ✓ Storage limitations change



2: MMDB vs Classical Databases

Video platform:

- Video playlist:
 - Semi structured data model
 - XPath ... structured matching and limited indexing
 - Embedded links to video instances
 - Link unstructured data models
 - Provide structure and maintain hyperlinks
 - No direct connection to the data



2: MMDB vs Classical Databases

Video platform:

- Video recommender:
 - Semi structured data model
 - Semantic search ... fuzzy matching
 - Embedded links to video instances
 - ∞ Link unstructured data models
 - ∞ Provide structure and maintain hyperlinks
 - ∞ No direct connection to the data
- Result structure changes with underlying content

